

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458505.75457 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.045
Transit depth (Rp/Rs)^2	2.31 +/- 1.36 [%]
Orbital Inclination (inc)	86.58 +/- 2.86
Airmass coefficient 1 (a1)	0.029 +/- 0.017
Airmass coefficient 2 (a2)	3.0 +/- 0.23
Scatter in the residuals of the lightcurve fit is	37.85 %
Best Comparison Star	None
Optimal Aperture	5.84
Optimal Annulus	10.04
Transit Duration (day)	0.1327 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.045 vs 0.1178 (deviation 0.19σ)
Duration fit vs published	0.1327 d vs 0.125 d (deviation 0.3σ)
Mid-time O–C	-1.7 min
Depth SNR	0.8
Residual scatter	37.85 %

Light curve

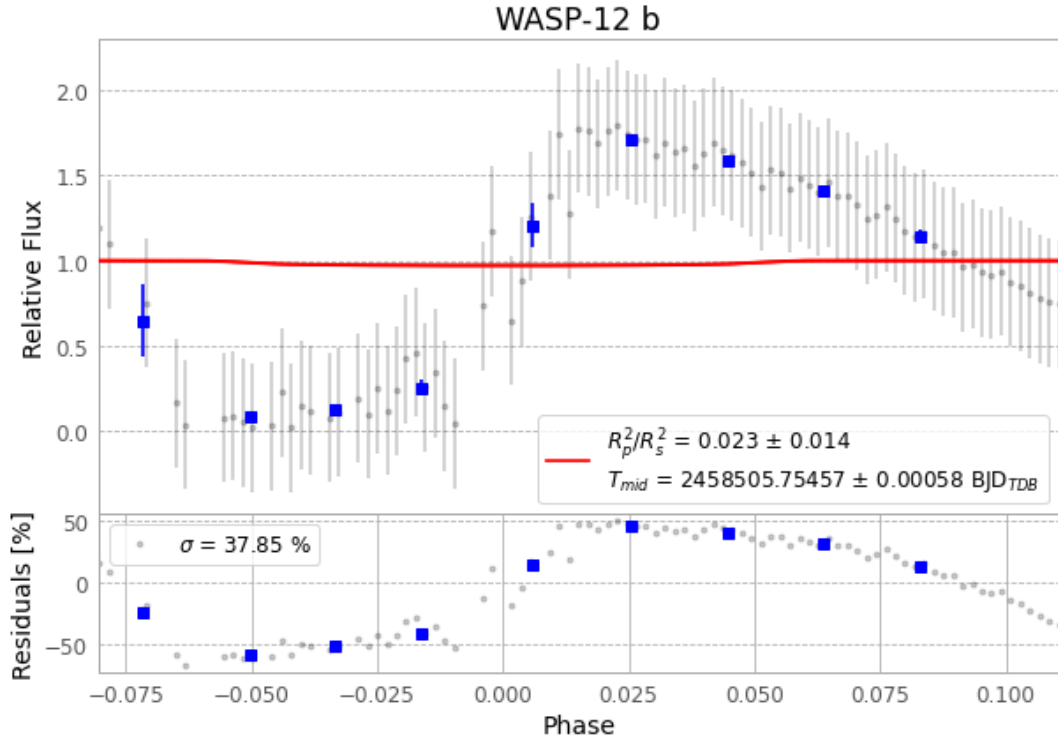


Figure 1 — FinalLightCurve_WASP-12 b_2019-01-21.png (SHA-256 281ba2be0b4b69e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [439, 295], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ca53528baf67a65e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-12190122035112.FITS → WASP-12190122085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190122.log (e372ffaec028...), FinalLightCurve_WASP-12 b_2019-01-21.png (281ba2be0b4b...), FinalLightCurve_WASP-12 b_2019-01-21.csv (f56f43e4d5d1...)

Compute resources

Wall time 270.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)